

Phase 1 Proposal: Stack Overflow Developer Q&A Lakehouse

Course Data Analysis and Visualization | Date 25 September 2026
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Repository <https://github.com/Suf-7/stackoverflow-lakehouse>

1. Domain and Source Identification

Concept. Stack Overflow is the largest public Q&A site for software developers. Its history shows which technologies developers struggle with, how quickly the community helps, and how that has changed since AI coding assistants arrived. Questions still on the site per year fell from 1.88 million in 2018 to 110 thousand in 2025 (Figure 1). We will track questions and answers for 18 data-and-analytics tags from 1 January 2018, which gives a five-year baseline before ChatGPT was released.

Tracked tags: pandas, numpy, python-polars, duckdb, apache-spark, pyspark, databricks, scikit-learn, tensorflow, pytorch, matplotlib, powerbi, tableau-api, airflow, dbt, snowflake-cloud-data-platform, sql, r. Names were validated against the API and are configurable in config/pipeline_config.json.

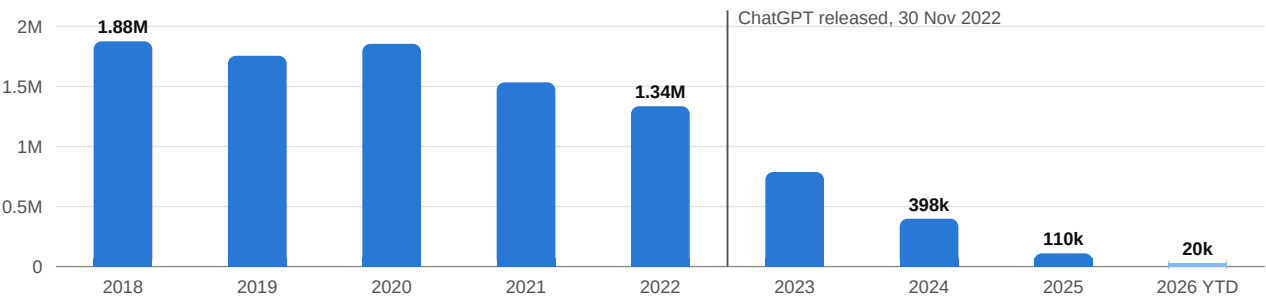


Figure 1. Questions on Stack Overflow per year, excluding deleted posts. Counted with the Stack Exchange API on 25 Sep 2026; 2026 is year to date.

Data source. Stack Exchange API v2.3 (<https://api.stackexchange.com/docs>), site [stackoverflow](https://stackoverflow.com). Public data needs no login. The quota is 300 requests per day without a key and 10,000 with a free stackapps.com key, which we will use for the full load. Content is licensed CC BY-SA 4.0 and will be attributed on the dashboard.

Purpose	Endpoint	Key parameters
Full load of questions	GET /questions	tagged, fromdate, todate, sort=creation, filter=withbody
Incremental changes	GET /questions	tagged, sort=activity, min=<watermark>
Answers	GET /questions/{ids}/answers	Up to 100 question ids per call
Deletions and metric refresh	GET /questions/{ids}	Up to 100 ids per call; missing ids are deleted

Ingestion pattern

Load	How it works
Full load (once)	For each tag, one calendar month per request window from 2018-01-01. Questions are de-duplicated on question_id, then answers are fetched for every answered question. Month windows make the job resumable.
Incremental (daily)	sort=activity&min=<watermark> returns new questions and questions whose last_activity_date moved: new answers, edits and closures. The watermark is the latest activity seen, minus a one-hour overlap.
Deletes and metrics (weekly)	The API never returns deleted posts, so known ids from the last 12 months are re-requested and missing ids become deletion records. The same call refreshes score and view counts, which do not change last_activity_date. Tested: a non-existent id was correctly flagged as deleted.

2. Data Samples and Volume

Samples were pulled live on 25 Sep 2026 with the repository scripts. Each batch is JSON Lines, one raw API record per line, plus a `_manifest.json` with batch id, extraction time, window, record counts and API usage.

Sample	Contents	Records	Size
Full load data/samples/full_load	Questions created 1 Apr to 30 Jun 2025 for all 18 tags, with bodies, and their answers	2,738 questions 3,384 answers	16.6 MB
Incremental data/samples/incremental	24 hours of activity; deletion check of all 2,738 full-load ids; metric snapshots for 100 ids	35 questions (2 new, 33 updated) 59 answers, 0 deletions 100 snapshots	0.2 MB

Load	Estimated volume	Basis
Full load, 2018 to date	~1.07M questions and ~1.3 to 1.6M answers ~6.5 to 7 GB raw JSON, ~1.8 GB gzipped, ~1.5 to 2 GB as Delta ~30,000 API calls, about 3 days with a key	Exact per-tag counts (1,164,064 tag matches) divided by 1.087 tracked tags per question; 3.3 KB per question and 2.3 KB per answer measured on the sample
Daily incremental	~35 changed questions and ~60 answers, ~180 KB	Measured 24-hour sample
Weekly reconciliation	A few thousand recent ids, ~1 MB of snapshots	Last 12 months of tracked questions
Monthly total	~6 MB	Daily rate times 30

The full load, over 2 million semi-structured records with HTML bodies and a many-to-many tag relationship, justifies Spark. The small incremental volume is a genuine finding about Stack Overflow today. It keeps daily runs cheap while still exercising upserts, deletions and changing metrics.

3. Security and Compliance

All content is public, but it still contains personal data. We profiled every field and scanned all bodies with regular expressions (scripts/profile_samples.py). No financial, payment or special-category data is present.

Data	Finding in the samples	Handling before Silver
owner.display_name, owner.profile_image, owner.link	Direct identifiers: often real names, personal Facebook or Google photo URLs, profile links	Drop
owner.user_id, owner.account_id	Pseudonymous ids that link a person across posts and sites	Hash with salted SHA-256 into user_key; salt kept in a Databricks secret scope
Title and body free text	Emails in 34 posts (about half placeholders), public-looking IPs, credential-like strings in 29 posts, connection strings in 16 posts	Derive and mask: main Silver keeps only features such as length and code-block count; a restricted table holds a copy with emails, IPs and secrets replaced by tokens
Reputation, user type, question metadata	Not identifying on their own	Keep

- Bronze keeps the raw payload for replay and is restricted to the two team members through Unity Catalog; the dashboard only reads Gold.
- Deletions on Stack Overflow propagate as soft deletes in Silver and are excluded from Gold.
- API keys and salts come from secrets, never from code. Only small samples are committed; full-size data folders are in .gitignore.

4. High-Level Medallion Data Modeling

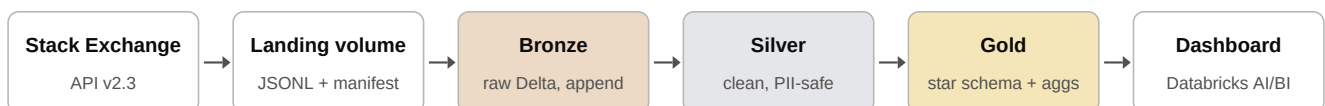


Figure 2. Planned pipeline. Full load once, incremental daily, deletion and metric reconciliation weekly, orchestrated with Lakeflow Jobs.

Bronze. Auto Loader appends landed JSONL into bronze.questions_raw, answers_raw, deleted_questions_raw and question_snapshots_raw, adding source file, batch id and ingestion time. A rescued-data column catches schema drift, which is real: optional fields such as migrated_from and protected_date appear on a few records.

Bronze to Silver cleaning, each driven by what the samples showed:

- Cast Unix epoch integers to UTC timestamps, and decode HTML entities in titles (245 of 2,738 titles contain them).
- De-duplicate on question_id and answer_id, then MERGE, updating only when the incoming last_activity_date is newer.
- Apply the PII rules; give posts from deleted users (39 in the sample) a null user_key.
- Derive body features, normalise close reasons, apply soft deletes, and separate *answered* from *accepted*: the API's is_answered is true for 66% of questions although 83.5% have an answer.
- Run quality checks: non-null keys, creation before last activity, every answer has a parent question.

Silver table	Grain	Main columns
silver.questions	Question, latest version	created_at, last_activity_at, closed_at, closed_reason, title, score, view_count, answer_count, accepted_answer_id, owner_user_key, body features, is_deleted
silver.question_tags	Question and tag	question_id, tag, is_tracked_tag
silver.answers	Answer, latest version	answer_id, question_id, created_at, score, is_accepted, owner_user_key, body features, is_deleted
silver.users	Pseudonymous user	user_key, user_type, latest_reputation, first_seen_at
silver.question_snapshots	Question and snapshot date	score, view_count, answer_count

Gold table	Type and grain	Purpose
fact_question	Fact, one row per question	Score, views, hours to first and accepted answer, closed flag
fact_answer	Fact, one row per answer	Score, accepted flag, hours after question, first-answer flag
bridge_question_tag	Bridge, question and tag	Resolves the many-to-many question-tag relationship
dim_tag, dim_date, dim_user	Dimensions	Tag with technology category; calendar with a before/after ChatGPT flag; hashed user with reputation band and cohort month
agg_tag_month	Aggregate, tag and month	Questions, share of tracked questions, % unanswered, median and p90 hours to first answer, % closed
agg_tag_pairs, agg_user_cohort	Aggregates	Tag co-occurrence counts; user retention by first-post month

5. Business Intelligence and Dashboards

Tool: Databricks AI/BI Dashboards, included in the Free Edition, reading Gold tables directly in the browser.

Business questions. (1) How much has developer Q&A declined since late 2022, and which technologies were hit hardest? (2) Where do developers wait longest for help? (3) Which technologies are gaining or losing share of developer attention?

- **AI-era trend:** monthly questions per technology category since 2018 with a marker at 30 Nov 2022, plus bars ranking tags by change from 2022 to 2025.
- **Response time:** median and 90th-percentile hours to first answer and % never answered, by tag. The sample already shows a large gap (table below).
- **Momentum quadrant:** each tag plotted by question volume against year-over-year change in share.
- **KPI tiles:** questions this month against the same month in 2022, % answered within 24 hours, median hours to first answer. Stretch goals: tag co-occurrence heatmap, retention cohorts, and a Spark MLlib model predicting whether a question gets answered.

Tag (Apr to Jun 2025 sample)	Questions	Median hours to first answer	Never answered
pandas	285	0.9	7.7%
sql	540	1.3	9.3%
r	883	1.5	15.2%
powerbi	159	10.6	27.7%
pyspark	131	18.9	26.0%
apache-spark	139	29.1	32.4%

6. Engineering Setup and FinOps

Version control. The repository holds the ingestion code (ingestion/), configuration, profiling script, samples with manifests, volume estimates and this proposal. Notebooks for Bronze, Silver and Gold follow in Phases 2 and 3.

Infrastructure. Databricks Free Edition: serverless compute, Unity Catalog volumes and tables, Delta Lake, Lakeflow Jobs. The ingestion scripts use only the Python standard library. If Free Edition compute cannot reach the API, ingestion will run as a scheduled GitHub Actions workflow that uploads each batch to a Unity Catalog volume with the Databricks CLI.

Risk	Control	
Daily compute allowance	Develop and test on the 17 MB repository samples; run the full load once; one small daily job; no always-on compute	
Storage	Land raw files gzipped (~1.8 GB); keep Delta tables after Bronze is verified; run OPTIMIZE once after the full load	
API quota	Free key for 10,000 requests a day; the client honours server backoff and stops at a quota reserve; the full load resumes by month window	
Wasted reruns	Every batch has a manifest and batch id, so failed runs replay from Bronze instead of calling the API again	
Phase	Due	Planned scope (to be adjusted when requirements are published)
2	10 Oct 2026	Full load into Bronze; Silver cleaning with PII rules; daily incremental job with MERGE and deletions
3	24 Oct 2026	Gold star schema and aggregates; dashboard; data-quality report